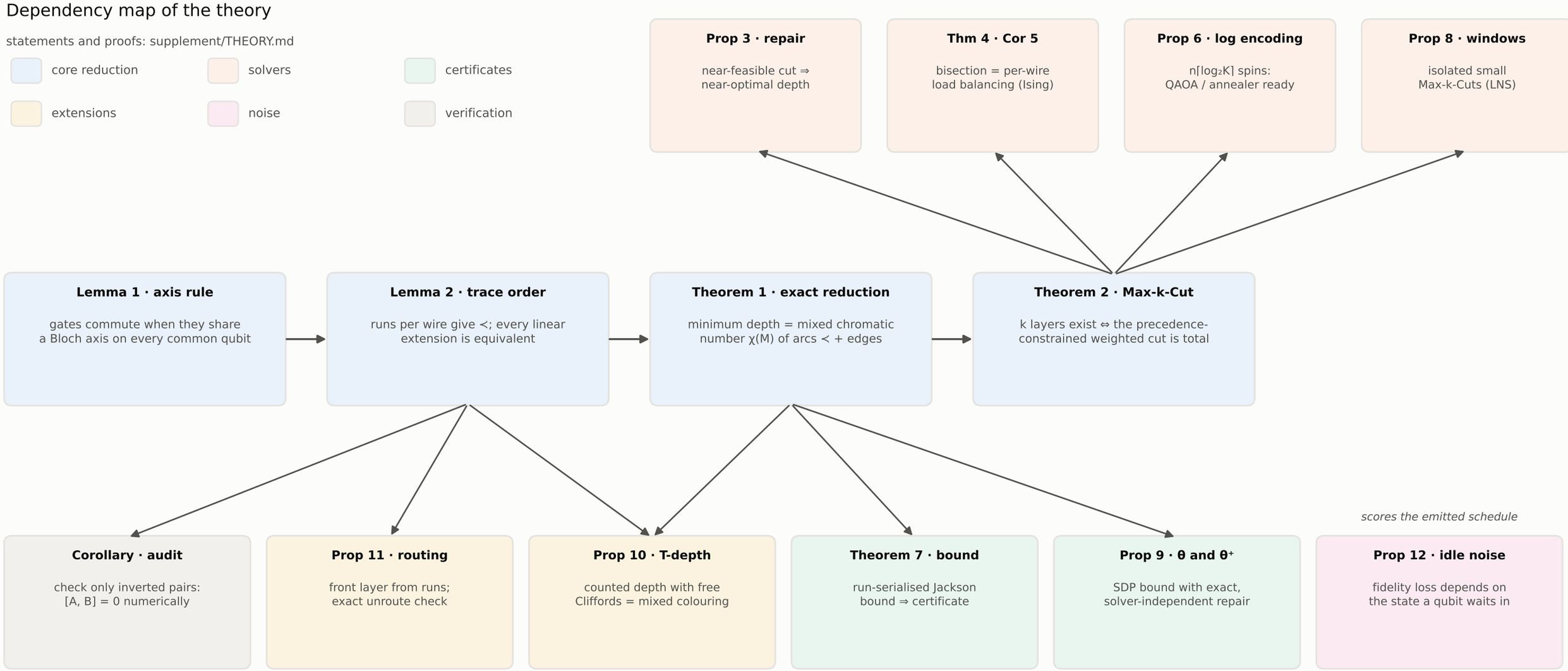
Dependency map of the theory
statements and proofs: supplement/THEORY.md
core reduction
solvers
certificates
extensions
noise
verification
Prop 3 · repair
near-feasible cut ⇒
near-optimal depth
Thm 4 · Cor 5
bisection = per-wire
load balancing (Ising)
Prop 6 · log encoding
n⌈log₂K⌉ spins:
QAOA / annealer ready
Prop 8 · windows
isolated small
Max-k-Cuts (LNS)
Lemma 1 · axis rule
gates commute when they share
a Bloch axis on every common qubit
Lemma 2 · trace order
runs per wire give ≺; every linear
extension is equivalent
Theorem 1 · exact reduction
minimum depth = mixed chromatic
number χ(M) of arcs ≺ + edges
Theorem 2 · Max-k-Cut
k layers exist ⇔ the precedence-
constrained weighted cut is total
Corollary · audit
check only inverted pairs:
[A, B] = 0 numerically
Prop 11 · routing
front layer from runs;
exact unroute check
Prop 10 · T-depth
counted depth with free
Cliffords = mixed colouring
Theorem 7 · bound
run-serialised Jackson
bound ⇒ certificate
Prop 9 · θ and θ⁺
SDP bound with exact,
solver-independent repair
scores the emitted schedule
Prop 12 · idle noise
fidelity loss depends on
the state a qubit waits in